

10.8 Summary and Notes

10.8.1 Summary

Idea.

1. Floating-point arithmetic commits a relative error of at most $u = 2^{-53}$ per operation (Definition 10.1.1). Inner products are backward stable (Proposition 10.1.2), but forming $\mathbf{X}^\top \mathbf{X}$ destroys the information in singular values below about $u^{1/2}\sigma_1$ (Proposition 10.1.3, Example (A cross-product matrix that rounds to a singular one)).
2. The Cholesky factorization of the augmented cross-product matrix gives $\hat{\boldsymbol{\beta}}$, the SSE, the sequential sums of squares, the leverages and the standard errors in $np^2 + p^3/3$ flops (Theorem 10.2.1, Proposition 10.2.2), accurately enough when the column-scaled condition number is modest. Sweeping fits nested models (Proposition 10.2.3); centring needs two passes or updating (Proposition 10.2.4).
3. Householder QR reduces $\mathbf{X}$ to a triangle by reflections in $2np^2 - 2p^3/3$ flops and is backward stable with columnwise errors (Theorem 10.3.2, Proposition 10.3.3). Givens rotations work one row at a time, for streaming and sparse data (Theorem 10.3.4).
4. The SVD gives every least squares solution, the minimum-norm one, and the precision σ/σ_i of each combination $\mathbf{v}_i^\top \boldsymbol{\beta}$ (Theorem 10.4.1). Singular values are perturbed by at most $\|\mathbf{E}\|_2$ (Proposition 10.4.2), so rank is meaningful only relative to a tolerance (Definition 10.4.1). Truncated SVD trades bias for variance (Proposition 10.4.3).
5. The least squares solution has sensitivity $\kappa + \kappa^2\eta$, where η measures the residual relative to the fit; residuals have sensitivity κ only (Theorem 10.5.1). The normal equations add κ^2 regardless. Equilibrating the columns is nearly optimal among scalings (Proposition 10.5.2, Example (Longley's data, exactly)).
6. A triangular factor can be updated or downdated by one row in $O(p^2)$ (Theorem 10.6.1), and by one column in $O(np)$ (Proposition 10.6.3). All leave-one-out residuals, coefficients and sums of squares follow from a single fit (Proposition 10.6.2).
7. QR with column pivoting orders the columns by how much each adds to the others, and a small trailing pivot certifies a small singular value (Theorem 10.7.2). Basic and minimum-norm solutions agree on everything estimable (Example (Firm effects in investment data)); the tolerance should reflect the accuracy of the data (Example (A nearly aliased column)).

10.8.2 Notes and sources

Source coverage. The chapter covers Seber and Lee (2003, chapter 11) on solving the normal equations, QR by Gram–Schmidt, Householder and Givens, the SVD, centring and leverages (sections 11.2–11.4, 11.7 and 11.10), with parts of sections 11.6, 11.8 and 11.9 on updating, comparing methods and rank deficiency.

Floating point and stability. The standard model and the γ_n notation follow Higham (2002),

whose chapters 3, 10, 19 and 20 are the definitive treatment of inner products, Cholesky, QR and least squares. Goldberg (1991) is a readable introduction to IEEE arithmetic, now specified by IEEE Std 754-2019. The distinction between a problem's condition and an algorithm's backward stability goes back to Wilkinson (1963). Trefethen and Bau (1997) give a short modern account. Läuchli (1961) introduced the matrix of Example (A cross-product matrix that rounds to a singular one) to show the loss in forming normal equations.

Algorithms. Householder (1958) introduced the reflections and Golub (1965) applied them to least squares; Givens (1958) used plane rotations. Gentleman (1973) developed square-root-free rotations and the row-wise algorithm, turned into widely used routines by Miller (1992). Björck (1967) analysed modified Gram–Schmidt, and Björck and Paige (1992) explained its good behaviour for least squares. The SVD algorithm is due to Golub and Kahan (1965) and Golub and Reinsch (1970), column pivoting to Businger and Golub (1965), and the counterexample of Example (Kahan's matrix) to Kahan (1966). Chan (1987) gave the first rank-revealing QR algorithm, Hong and Pan (1992) proved that factorizations with polynomial bounds exist, and Gu and Eisenstat (1996) showed how to compute them efficiently. Goodnight (1979) describes the sweep operator for statisticians. Welford (1962) gave the updating recursion, and Chan, Golub and LeVeque (1983) compare variance algorithms. Downdating appears in LINPACK (Dongarra et al. 1979) and was analysed by Stewart (1979). The standard monographs are Björck (1996), Lawson and Hanson (1974) and Golub and Van Loan (2013); Thisted (1988) and Kennedy and Gentle (1980) take the statistician's side.

Perturbation theory. Golub and Wilkinson (1966) first observed that the least squares solution can be sensitive to κ^2 when the residual is large. Wedin (1973) gave sharp perturbation bounds for pseudoinverses and least squares. The equilibration result of Proposition 10.5.2 is from van der Sluis (1969). The inequality of Proposition 10.4.2 is the singular-value form of Weyl's (1912) inequality for eigenvalues. Beaton, Rubin and Barone (1976) argued that errors from rounding the data should be the yardstick for computational accuracy, the point made at the end of [Section 10.5](#).

Statistics. Recursive least squares via the Sherman–Morrison formula goes back to Plackett (1950). The PRESS statistic is due to Allen (1974). The recursive residuals of Exercise (C1) are those of Brown, Durbin and Evans (1975). Deletion diagnostics are developed in [Chapter 20](#), collinearity in [Chapter 26](#), and ridge and principal components regression in [Chapter 27](#).

10.8.3 References

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